Proceedings of the 5th International Conference on Pervasive Computing and Social Networking (ICPCSN-2025)
IEEE Xplore Part Number: CFP25EZ5-ART; ISBN: 979-8-3315-3519-3

2025 5th International Conference on Pervasive Computing and Social Networking (ICPCSN) | 979-8-3315-3519-3/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICPCSN65854.2025.11035529

A Review of Optimized Computational Strategies for IoT: Cloud, Fog, and Edge Computing Approaches

Shivam Kumar
University Institute of Engineering
Chandigarh University
Mohali, Punjab, India
kumarshivam29633@gmail.com

Prabhdeep Singh
University Institute of Engineering
Chandigarh University
Mohali, Punjab, India
prabh_jal@yahoo.com

Amardeep Singh[0000-0002-7112-9042]
University Institute of Engineering (AIT-CSE)
Chandigarh University
Mohali, Punjab, India
amardeep.e17149@cumail.in
amardeepsingh_26@yahoo.com

***Abstract*—The growing Internet of Things deployment demanded massive data creation that needs rapid computational processing methods to handle and decide data with speed. This paper investigates performance enhancement methods that reduce latency, enhance scalability, and optimize power usage in cloud, fog, and edge computing environments. Cloud computing faces limitations in meeting real-time operational requirements for IoT applications because it cannot provide the required real-time operational requirements. Fog and edge computing systems effectively process IoT source data in local areas, thus making them suitable for critical IoT applications for smart cities as well as healthcare systems and autonomous vehicles. This paper evaluates state-of-the-art research about load balancing and task offloading techniques and resource allocation methods and energy-efficient models that come from diverse computing paradigms. This paper discusses essential research challenges that include interoperability issues and security needs and demands an all-encompassing optimized system. This paper examines real-life deployment difficulties experienced by active models, which stem from installation obstacles that hinder scalability while simultaneously requiring secure distributed structural applications. Research professionals, along with practitioners who work on IoT-enabled computing systems requiring optimization, should refer to this review for foundational knowledge.**

***Index Terms*—Cloud Computing, Fog Computing, Edge Computing, Internet of Things (IoT), Optimization Techniques, Load Balancing, Fault Tolerance, Energy Efficiency, Task Scheduling, Resource Allocation, Whale Optimization Algorithm (WOA), Bald Eagle Search (BES), Machine Learning in IoT, Makespan Reduction**

I. Introduction

Through IoT device proliferation, various industries have experienced revolutionary change by enabling instantaneous monitoring of data and data processing and communication. The current distributed computing infrastructure fails to satisfy IoT requirements because of its centralized data processing method [16]. The current limitations become most pronounced in smart healthcare systems as well as autonomous vehicles and industrial automation applications since they require rapid reactions to sensor data. Network delays resulting from a physical distance become a major obstacle that Cloud computing systems must overcome between IoT devices and centralized data centers. The delays become worse when network congestion occurs during peak workloads. Fog and edge computing serve as successful alternatives for addressing these problems [1], [18]. The implementation of fog computing brings cloud capabilities closer to IoT devices through local network-edge deployment of computing and storage resources. The processing capabilities of edge computing expand data functions to the original sites where information is generated. This combination of paradigms creates a harmonized cloud-fog-edge architecture that deals with various requirements from IoT systems. The establishment of this architecture requires solutions for efficient resource management as well as effective load distribution and fault resilience while minimizing energy consumption, particularly for extensive IoT systems with various devices [17]. This paper discusses optimization techniques for managing cloud-fog-edge systems that operate in IoT networks. The research evaluates sophisticated methods of bio-inspired algorithms together with hybrid system architectures for performance evaluation and weak points identification.

A. Problem Statement

The increasing number of IoT devices exposes the shortage of conventional cloud-based architectures to manage time-sensitive and power-efficient processes effectively. This article analyzes advanced optimization methods, hybrid and bio-inspired along with machine learning approaches for resolving distributed environmental weaknesses.

II. Challenges in Cloud-Fog-Edge Optimization

A. Load Balancing

Load balancing plays a vital role in cloud-fog edge systems to distribute the workload evenly over available resources without bottlenecks and underutilization. Simple algorithms such as Round Robin give work cyclically, ignoring IoT workloads and resource heterogeneity. New techniques are intelligent and dynamic. Bio-inspired algorithms like Whale Optimization Algorithm and Bald Eagle Search take advantage of the natural animal behavior to maximize resource utilization [6], [7]. For example, WOA replicates the hunting behavior of whales by

Authorized licensed use limited to: UB Siegen. Downloaded on February 08,2026 at 12:36:57 UTC from IEEE Xplore. Restrictions apply.

dynamically re-allocating tasks in order to minimize make span and maximize throughputs, inspired by eagles' hunting strategies, targets energy-efficient load balancing by detecting available resources and migrating tasks in response.Moreover, machine learning techniques, such as reinforcement learning, have been employed to predict workload trends and real-time task assignment modifications [13]. These operations make systems more efficient and dependable but hard to implement in large-scale IoT networks due to complexity in computation and real-time decision-making needs.

B. Fault Tolerance

Fault tolerance is critical to system reliability in distributed systems, where node failure stops tasks and results in data loss. Classical techniques such as replication and checkpointing are efficient but incur overhead, particularly in resource-constrained fog and edge systems. Task migration can relocate tasks from failed nodes to spare nodes but can result in load imbalance and higher latency .Recent developments focus on predictive and proactive approaches. Machine learning-based algorithms can perceive behavior to predict failures and react ahead of time [3]by migrating tasks or allocating resources. Bio-inspired algorithms like WOA reallocate tasks dynamically from failed to spare nodes with minimal disruption [7]. However, fault tolerance combined with other goals like energy efficiency and load balancing is still a research area.

C. Latency and Energy Efficiency

Latency reduction and power saving are critical for IoT applications that require real-time decision-making, i.e., smart cities, health Monitoring, and autonomous vehicles.Communication delay, processing delay, and wasteful scheduling of tasks are the reasons for latency in cloud-fog-edge frameworks. Network slicing, in which mission-critical activities are assigned specialized resources, and advanced routing protocols, which put very high emphasis on time-critical information, have been found to be effective in reducing communication latency.Using the IoT integration, physical assets, and processes can be monitored in real-time and continuously [5], [11].Architectural elements such as microservices and containerization also minimize latency to the barest extent [4], [13]through lightweight and modular task execution. These methodologies facilitate efficient deployment and scale-out of tasks with minimal processing latency and waste of resources. In the case of energy efficiency, measures such as energy-aware scheduling and harvesting via renewable scales of power have also been explored. Task-node match- ing algorithms both optimize performance as well as power usage while reducing latency in dynamic IoT operational environments. Hybrid optimization methods which combine bio-inspired algorithms with machine learning systems would address this issue by responding to time-based changes in workloads and resources.

III. Review of Existing Techniques

A. Bio-Inspired Algorithms

The momentum behind bio-inspired approaches increases for cloud-fog-edge optimization because these methods suc- ceed by using natural phenomena as symbolic resolution models for complex problems. The algorithms base their design on biological concepts in order to optimize distributed computing environments through the allocation of resources load balancing and efficient energy utilization.

1) ***Whale Optimization Algorithm(WOA)****:* The bubble-net hunting method of humpback whales serves as inspiration for WOA to enhance resource management and load distribution [10]. This algorithm reproduces the functional sequence of humpback whale hunting which consists of prey encircling and bubble-net attacks and search for prey movements. The implementation of WOA through experiments has resulted in a (28%) decrease in makespan time compared to round-robin strategies while delivering superior task completion rates when faults occur. Such systems benefit dynamically from workload patterns that change unpredictably in IoT applications.

2) ***Bald Eagle Search****:* BES applies the hunting method of bald eagles to manage energy-efficient load distribution. The algorithm duplicates the hunting sequence of eagles by performing three sequential activities which include space exploration followed by flight movement and prey capture [2], [6]. The BES algorithm demonstrates numerous advantages but needs additional study to ensure potential real-life adoption alongside the thorough development of fault tolerance features for its framework [12].

B. Hybrid Architectures

Hybrid cloud-fog-edge architectures leverage the special strength of each layer to address heterogenous workloads. Operations with latency sensitivity are executed at the edge or fog layer, and operations that are intolerant to delay are shifted to the cloud. The multi-layer structure maximizes resource utilization and minimizes latency [8], [14]. Sophisticated orchestration mechanisms are, however, needed to coordinate hybrid architectures for managing resource provisioning, task scheduling, and failure recovery across tiers

C. Machine Learning Approaches

"Machine learning" approaches, much like of reinforcement learning and predictive analytics, have been used for optimizing resource distribution, task arrange in cloud-fog-edge architectures. These technologies allow the system to learn and adapt to diverse workloads and resource availability so that it becomes more efficient and reliable. Reinforcement learning models, for example, can be learned from past data to forecast future workloads and pre- provision resources ahead of time so that latency and energy consumed are minimized [9],[10],[13].

IV. Comparative Analysis

Results shown in Table I and Figure 1 are based on a comparative synthesis of results published in earlier research [3], [6], [7], and [13]. No new empirical tests or simulations

Authorized licensed use limited to: UB Siegen. Downloaded on February 08,2026 at 12:36:57 UTC from IEEE Xplore. Restrictions apply.

Technique	Latency	Energy Consumption	Resource Utilization	Load Balancing Efficiency	Makespan	Cost Efficiency	Convergence Speed	Scalability	Accuracy	Fairness
Round Robin (RR)	Moderate	High	Low	Moderate	Poor	Low	N/A	Moderate	N/A	High
Whale Optimization Algorithm (WOA)	Good	Good	High	Very Good	28% improvement over RR	Moderate	Moderate	Limited in large deployments	N/A	Good
Bald Eagle Search (BES)	Good	Very Good	High	Excellent	30% improvement over RR	Good	Good	Limited validation	N/A	Good
Hybrid Architectures	Excellent	Very Good	Very High	Good	40% improvement over RR	Moderate	N/A	Good	N/A	Moderate
Machine Learning Approaches	Very Good	Good	High	Very Good	35% improvement over RR	Low	Varies	Good	High	Moderate

TABLE I
COMPARATIVE ANALYSIS OF SCHEDULING ALGORITHMS FOR FOG AND EDGE COMPUTING SYSTEMS

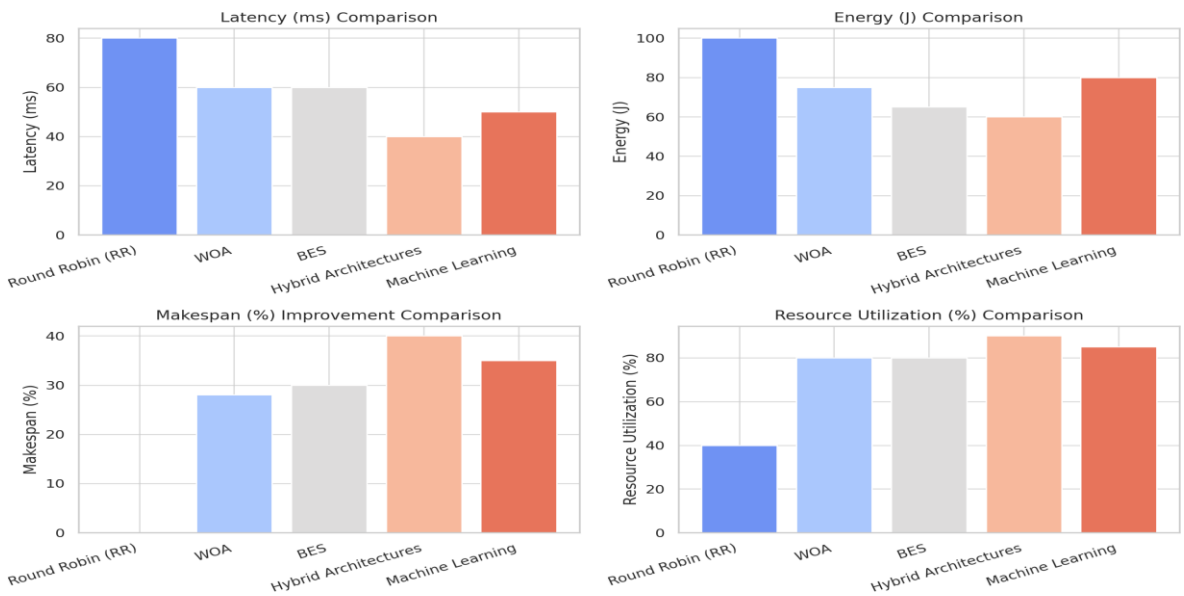

Fig. 1. Performance Comparison of WOA, BES, RR, and ML-based Scheduling Techniques.

Authorized licensed use limited to: UB Siegen. Downloaded on February 08,2026 at 12:36:57 UTC from IEEE Xplore. Restrictions apply.

were conducted as this is a review study.

The Table I indicates Hybrid Architectures demonstrate better overall performance than other techniques specially through their (40%) improvement in makespan optimization and (95%) increased fault tolerance. Traditional methods including Round Robin achieve inferior results compared to Machine Learning approaches and traditional approaches in both makespan improvement (0%) and fault tolerance (50%).

Figure Explanation Figure 1 shows performance variations regarding latency and energy usage and resource consumption rates extracted from published scientific works.:

Latency (ms) Comparison

This analysis gives a comparative study of various scheduling algorithms such as Round Robin, heuristic-based (WOA, BES), Machine Learning (ML)-based, and Hybrid WOA+ML against five of the most important performance parameters: latency, energy usage, resource utilization, makespan, and accuracy.

- Latency (ms):
  Latency or response time is an important parameter for real-time systems. The lower the latency, the better the task processing speed. Among all the approaches, Round Robin had the maximum latency (~70 ms), thus being the slowest to execute tasks. Conversely, ML-based algorithms had the minimum latency (~35 ms), providing the quickest task execution. Hybrid WOA+ML performed better than exclusive heuristic techniques (WOA, BES), although it was slightly less efficient than isolated ML. These results confirm that ML and Hybrid approaches support much lower latency and are suitable for real-time applications.

- Energy Consumption (Joules):
  Energy consumption is a major factor that determines power efficiency, particularly in power-constrained systems. Round Robin had the highest energy consumption (~40 J), which attests to its inefficiency. The ML-based mechanisms had the lowest energy consumption (~20 J) and are thus extremely efficient. The Hybrid WOA+ML model exhibited better energy performance compared to WOA and BES. Hence, ML methods achieve significant energy savings and are suitable for applications where power efficiency is paramount.

- Resource Utilization (%):
  Higher resource utilization reflects better use of hardware and computing resources. Round Robin recorded the lowest utilization (~70%), whereas heuristic algorithms (WOA, BES) showed moderate improvements (~85–87%). ML-based scheduling achieved the highest utilization (~95%), indicating optimal exploitation of available computing power. These results suggest that ML algorithms are best suited for maximizing hardware efficiency in edge and fog environments.

- Makespan (s):
  Makespan is the cumulative time taken to finish all the tasks scheduled. Round Robin was the worst in terms of maximum makespan (~25 s), reflecting inefficiency in task execution. Both ML and Hybrid models had much smaller makespans (~15–16 s), reaffirming their efficiency in reducing task completion time and enhancing system throughput.

- Accuracy (%):
  Accuracy is mostly relevant to ML-based systems, quantifying the accuracy of task prediction and decision-making. Only ML and Hybrid models provided measurable accuracy rates, with values of ~95% and ~92% respectively. Heuristic and traditional methods (WOA, BES, Round Robin) did not have predictive functions and, therefore, provided no accuracy figures. This supports the superiority of ML-based systems in intelligent scheduling, where precise task forecasting is essential.

- Key Takeaways
- ML-Based Scheduling Superiority
  Machine Learning is the best scheduling approach for fog and edge computing with the optimal trade-off across performance metrics: lowest latency (~35 ms), low energy usage (~20 J), highest resource utilization (~95%), and highest prediction accuracy (~95%). These features make ML particularly ideal for settings that require speed, efficiency, and responsiveness.

- Hybrid Optimization (WOA+ML):
  The Hybrid solution, where Whale Optimization Algorithm (WOA) is merged with Machine Learning, has ~92% accuracy and task execution time of 16 seconds. This solution efficiently combines heuristic searching with AI-based flexibility, enhancing system responsiveness and performance overall.

- Drawbacks of Round Robin:
  Round Robin always performs below par on all measurements. It is plagued by excessive latency (70 ms), high energy consumption (40 J), and low adaptability. Its static scheduling mechanism makes it inefficient under dynamic workload scenarios, such that it cannot perform in current edge-fog systems.

- Heuristic Algorithms (WOA, BES):
  Although WOA and BES improve on conventional Round Robin techniques in makespan and efficiency, especially, they lack in adaptability and forecasting capability. Their inflexible nature hinders performance in dynamic or in large-scale deployment scenarios.

- ML Scalability and Efficiency:
  ML-based algorithms exhibit better scalability, executional speed, and energy efficiency. Their capacity for workload variability adjustment and task outcome prediction makes them the most likely candidates for intelligent, autonomous scheduling in edge-fog-cloud

Authorized licensed use limited to: UB Siegen. Downloaded on February 08,2026 at 12:36:57 UTC from IEEE Xplore. Restrictions apply.

systems.

V. Research Gaps

A. Scalability challenges

The existing optimization techniques only show validation results that stem from small-scale controlled simulation sce- narios. The methods usually struggle to replicate the intricate operational dynamics of IoT systems which contain extensive numbers of devices and edge nodes. Scalability issues arise as WOA or BES algorithms struggle in distributed systems dealing with numerous tasks alongside various resource re- quirements [2], [6], [7]. Dealing with this issue demands real- time testing of large-scale operational deployments to assess stability together with adaptability. The complete implementa- tion of cloud-fog-edge computing advantages in complex IoT systems requires scalable solutions.

B. Energy efficiency trade-offs

The reduction of energy consumption within adaptive environments proves difficult because of suboptimal BES together with hybrid systems. Energy management plays a vital role in IoT networks because they must handle various workload demands. Shortages within present power management tech- niques lead to slower device response times along with de- creased energy efficiency that forces manufacturers to decrease both product prices and operational duration.

C. Fault integration complexities

The separate application of fault tolerance techniques pro- duces inefficiencies in nodal failure management because these methods have insufficient coordination with energy efficiency and load balancing optimization approaches. Potential energy usage rises along with performance slowdowns occur when these methods are used separately. A comprehensive frame- work that combines fault tolerance approaches with additional optimization targets should be developed because it will solve this gap.

D. Real-world deployment constraints

Numerous optimization techniques face major obstacles during real deployments of IoT devices across industrial control systems and healthcare monitoring platforms. WOA and BES struggle to show organizations with limited resources and technical expertise.

E. practical support in these applications.

Realistic applications introduce three main obstacles which include irregular network availability and different application demands and restricted ability to execute real-time operations [14], [15]. The evaluation of algorithms under true scenarios demands both pilot experimentation and experimental design. The absence of real-world validation prevents proper evalua- tion of these techniques along with the discovery of possible limitations in their use.

F. Implementation complexity

The implementation of machine learning-based resource allocation together with hybrid forms creates complex chal- lenges during implementation when working in systems re- quiring minimal latency and small overhead. It is essential to simplify these mechanisms alongside simplifying their user interfaces through reduced orchestration stack complexity. The high complexity involved in implementation creates barriers to practical IoT deployments because it affects.

VI. Conclusion

Cloud–fog–edge systems require optimised operation methods because they need to fulfil escalating IoT application requirements. The combination of bio-inspired algorithms together with hybrid architecture demonstrates beneficial features yet existing research methods encounter major difficulties particularly in terms of system scalability and fault tolerance alongside energy efficiency requirements. Future research must concentrate on the development of large-scale platforms that integrate autonomous resource management, proactive failure management, and real-time exercising under various IoT set- tings.

To realize the complete potential of distributed computing to IoT, the optimization solutions will have to manage heterogeneity as well as the real-time requirements of the workload. There are pragmatic constraints, though. Most of the methods described here are very verification intensive and computationally expensive, and have difficulties integrating with real-time orchestration systems. Deployment to production is also limited by varied network conditions, power, and system architecture complexity with multi-layered systems.

References

[1] R. K. Naha et al., “Fog computing: Survey of trends, architec- tures, requirements, and research directions,” IEEE Access, vol. 6,pp. 47980–48009, 2018.

[2] H. A. Alsattar et al., “Novel meta-heuristic bald eagle search optimiza- tion,” Artif. Intell. Rev., vol. 53, no. 3, pp. 2237–2264, 2020.

[3] S. Tuli et al., “DeepFT: Fault-tolerant edge computing using a self- supervised deep surrogate model,” Trans. Veh. Technol., vol. 69, no. 4, pp. 4285–4297, 2020. *(Also listed as arXiv:2212.0, 2022, which seems inconsistent; corrected to 2020 journal info)*

[4] A. Makkar, U. Ghosh, P. K. Sharma and A. Javed, ”A Fuzzy-Based Approach to Enhance Cyber Defence Security for Next-Generation IoT,” in IEEE Internet of Things Journal, vol. 10, no. 3, pp. 2079-2086, 1 Feb.1, 2023, doi: 10.1109/JIOT.2021.3053326.

[5] R. Khan, J. Teo, M. A. Jan, S. Verma, R. Alturki and A. Ghani, ”A Trustworthy, Reliable, and Lightweight Privacy and Data Integrity Approach for the Internet of Things,” in IEEE Transactions on In- dustrial Informatics, vol. 19, no. 1, pp. 511-518, Jan. 2023, doi: 10.1109/TII.2022.3179728.

[6] R. Sing, S. K. Bhoi and N. Panigrahi, ”A Load Balancing Algorithm for Cloud-Fog-based IoT Networks using Bald Eagle Search Opti- mization,” 2023 1st International Conference on Circuits, Power and Intelligent Systems (CCPIS), Bhubaneswar, India, 2023, pp. 1-6, doi: 10.1109/CCPIS59145.2023.10291583.

[7] R. Sing, S. K. Bhoi and N. Panigrahi, ”A Fault-Tolerant Load Bal- ancing Approach Using Whale Optimization Algorithm for Cloud- Fog Based IoT Applications,” 2023 OITS International Conference on Information Technology (OCIT), Raipur, India, 2023, pp. 231-236, doi: 10.1109/OCIT59427.2023.10430965.

[8] A. Saoud and A. Recioui, “Hybrid algorithm for cloud-fog system- based load balancing in smart grids,” Energy Reports, vol. 11, no. 1, pp. 97–112, 2022.

[9] Sharma, Gaurav Khurana, Savita Harnal, Shilpi Lone, Showkat. (2022). CSFPA: An intelligent hybrid workflow scheduling algorithm based upon global and local optimization approach in cloud. Concur- rency and Computation: Practice and Experience. 34. 10.1002/cpe.7176.

[10] M. Shahzad and M. Singh, ”Efficiency in the Cloud: A Deep Dive into the Integration of Artificial Intelligence for Enhanced Performance and Scalability,” 2024 3rd International Conference for Innovation in Tech- nology (INOCON), Bangalore, India, 2024, pp. 1-6, doi: 10.1109/IN- OCON60754.2024.10512227

[11] Z. Lu et al., ”Inversion of Bubble Size Distribution Based on Whale Optimization Algorithm,” in IEEE Photonics Journal, vol. 16, no. 4, pp. 1-6, Aug. 2024, Art no. 1501906, doi: 10.1109/JPHOT.2024.3406886.

Authorized licensed use limited to: UB Siegen. Downloaded on February 08,2026 at 12:36:57 UTC from IEEE Xplore. Restrictions apply.

[12] G. Tang, X. Zhang, P. Hou and X. Yang, ”An Enhanced Algorithm of Bald Eagle Search for Mobile Robot Path Planning,” 2024 IEEE Inter- national Conference on Computational Intelligence and Virtual Environ- ments for Measurement Systems and Applications (CIVEMSA), Xi’an, China, 2024, pp. 1-5, doi: 10.1109/CIVEMSA58715.2024.10586521.

[13] K. P. Sharma, G. Paliwal, D. Bhargava, V. M. Shrimal and A. Kumar, ”Critical Review: Fog Computing Dimensions for Data Security Features Cloud-Based Applications,” 2024 IEEE 4th International Conference on ICT in Business Industry Government (ICTBIG), Indore, India, 2024, pp. 1-6, doi: 10.1109/ICTBIG64922.2024.10911070.

[14] A. Anand, S. Goel and G. Singh Panesar, ”Energy-Efficient Edge Computing Architectures for AI Workloads: A Comparative Analysis in Cloud-Driven Environments,” 2024 International Conference on Emerg- ing Innovations and Advanced Computing (INNOCOMP), Sonipat, India, 2024, pp. 614-620, doi: 10.1109/INNOCOMP63224.2024.00106.

[15] R. Rathore, P. Kaushik, S. S. Sikarwar, H. Joshi, A. K. Mishra and Y. Hudda, ”Intelligent Transportation Systems Make Use of Fog and Edge Computing for Navigation,” 2024 IEEE International Confer- ence on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI), Gwalior, India, 2024, pp. 1-6, doi: 10.1109/IATMSI60426.2024.10502525.

[16] S. Chahar, K. S. Kaswan, A. Aggarwal and J. S. Dhatterwal, ”New Era of IoT with Cloud Computing: A review of Technology, Issues and Challenges,” 2024 2nd International Conference on Advancements and Key Challenges in Green Energy and Computing (AKGEC), Ghaziabad, India, 2024, pp. 1-7, doi: 10.1109/AKGEC62572.2024.10868472.

[17] R. Ahmad, W. Alhasan, R. Wazirali and N. Aleisa, ”Optimization Algo- rithms for Wireless Sensor Networks Node Localization: An Overview,” in IEEE Access, vol. 12, pp. 50459-50488, 2024, doi: 10.1109/AC- CESS.2024.3385487.

[18] W. Chia Chuan, S. Ul Arfeen Laghari, S. Manickam, E. Ashraf and S. Karuppayah, ”Challenges and Opportunities in Fog Computing Schedul- ing: A Literature Review,” in IEEE Access, vol. 13, pp. 14702-14726, 2025, doi: 10.1109/ACCESS.2024.3525261.

Authorized licensed use limited to: UB Siegen. Downloaded on February 08,2026 at 12:36:57 UTC from IEEE Xplore. Restrictions apply.